Regional Sea Surface Height Exploration and Analysis

https://github.com/arrathore/sea-level/

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ABSTRACT

This project will analyze long-term changes in global and regional sea levels using satellite altimetry data provided by NASA. Python-based data analysis tools will be used to retrieve sea level datasets and compute the global mean sea level over time. A primary objective will be to estimate the long-term linear trend in the sea level rise and investigate whether there is a measurable acceleration in the rate of increase. The analysis will help illustrate how satellite observations can be used to detect changes in the Earth's ocean system.

In addition to global analysis, the project will examine regional variations in sea level change. Sea level does not rise uniformly across the globe due to ocean circulation, temperature differences, and other factors. By dividing the dataset into geographic regions, the project will compare regional trends with the global average to identify areas experiencing faster or slower sea level change. Python visualization libraries will be used to create plots and a map showing these sea level trends. The visualizations will highlight the differences in sea level rise across different parts of the global ocean. Overall, the aim of the project is to demonstrate how large-scale datasets can be used and analyzed programmatically to reveal meaningful climate-related patterns and changes in ocean behavior.

1.0 INTRODUCTION

Though one may intuitively assume that the change in sea level across the global ocean follows a "bathtub model", where the height rises equally across all regions, the reality is not so simple. Different regions experience changes in sea surface height asymmetrically, for a variety of reasons. This project will take numerical data, and visualize these differences using plots and maps. I will then go on in the paper to explain the various reasons for these deviations.

NASA offers a number of useful datasets for analyzing sea level trends. NASA obtains this data via satellite altimetry which has been in use since the launch of Seasat in 1979. Seasat's objective was to "[demonstrate] nearly all-weather microwave surveillance of the world's oceans", that is, to allow for observation of the ocean from space in nearly any weather conditions, using microwave frequencies. [1] Satellite altimetry works by determining the satellite's altitude with respect to an Earth reference, and the distance between the satellite and the sea surface. By subtracting these two values, we can calculate the sea surface height.[2] This simple principle, when applied over the entire ocean surface, can provide a remarkably clear picture of both global and regional sea surface height.

These reference height calculations are made possible by radar and GPS, and can provide measurements of sea surface height accurate within 2–3 cm. These measurements can also help determine other information about the ocean, such as mapping the ocean floor. Because "the sea surface responds to changes in the gravitational attraction generated by topographic features on the ocean floor", it is possible to estimate the topology of the sea floor.[3]

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[1] Born, Seasat Mission Overview, 1.
[2] Stammer, *Satellite Altimetry*, 1–2.
[3] Martin, *An Introduction to Ocean Remote Sensing*, 362–365.

The project aims to write software to take this raw data and visualize it, making trends clear, and distinguishing global trends and regional variation. The first task, therefore, was to select datasets to work with. Several datasets were considered, but only two were finally selected:

- Global Mean Sea Level Trend from Integrated Multi-Mission Ocean Altimeters TOPEX/Poseidon, Jason-1, OSTM/Jason-2, Jason-3, and Sentinel-6 Version 5.2[4]
  (referred to hereafter as "*Global Mean Sea Level Trend*")
- NASA-SSH Simple Gridded Sea Surface Height from Standardized Reference Missions Only Version 1[5]
  (referred to hereafter as "*Simple Gridded Sea Surface Height*")

Global Mean Sea Level Trend was selected first, as it is a fairly simple dataset that we can use to get an idea of the global trend in sea level rise. It also allows for an easy introduction to working with the Python libraries for data processing and visualization. However, this dataset does not provide any specific geographical information for its measurements, and cannot be used to determine regional trends. *Simple Gridded Sea Surface Height* is the main dataset for this project. It contains "gridded" sea level data, meaning that data is marked by location. With this data, we can select any region and explore specific trends.

2.0 DATASET CALCULATIONS AND ANALYSIS

Both datasets were generated by a variety of satellites, including TOPEX/Poseidon, Jason-1, OSTM/Jason-2, Jason-3, and Sentinel-6A. The data retrieved from *Global Mean Sea Level Trend* spans from 1992 to 2026, and the data retrieved from *Simple Gridded Sea Surface Height* spans from 2014 to 2026. *Simple Gridded Sea Surface Height* is updated regularly, and it is technically

[4] NASA, Global Mean Sea Level Trend.
[5] NASA, Simple Gridded Sea Surface Height

possible to write a program that would be able to measure the changes in sea surface height as more data is collected.

2.1 Global Mean Sea Level Rise

Data analysis was performed in Python (all source code created for this project is made freely available at github.com/arrathore/sea-level/). First, *Global Mean Sea Level Trend* was used to create a figure of the general change in sea level, and the acceleration.

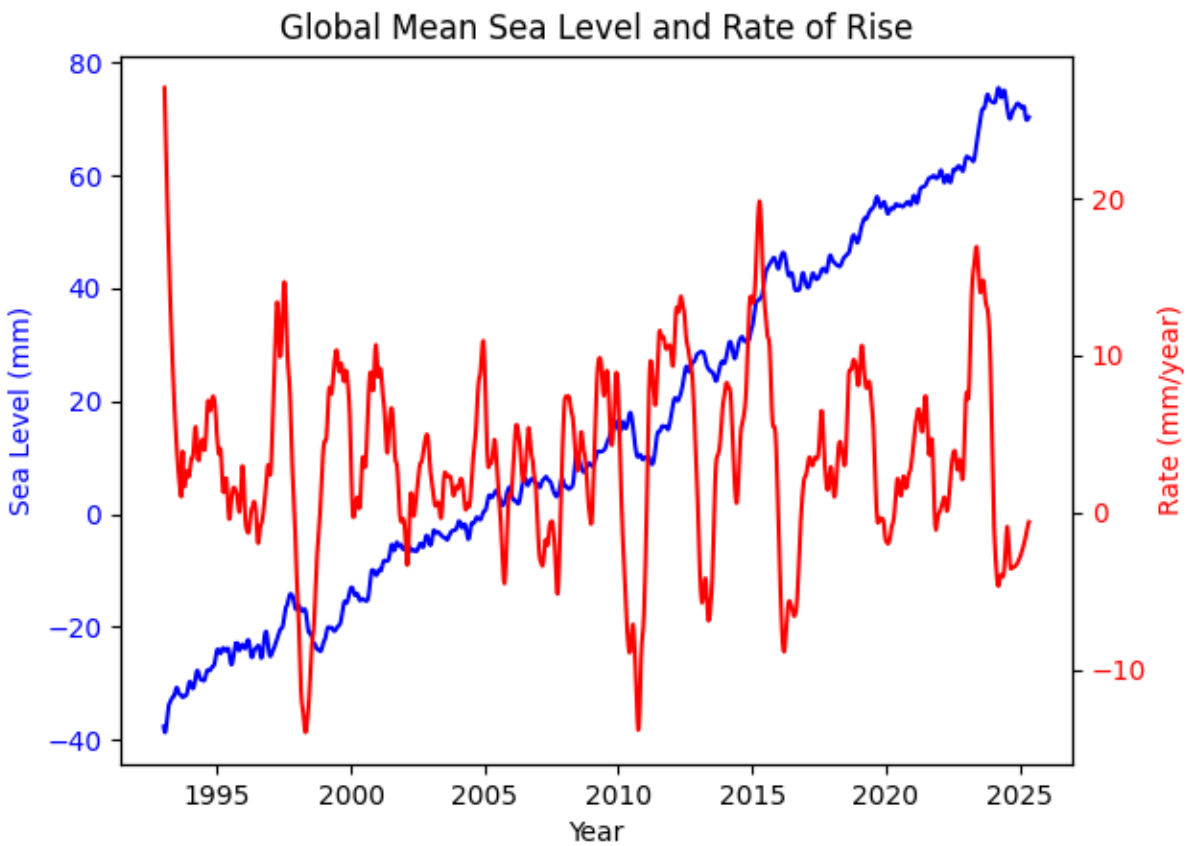

Figure 1: *A plot of global mean sea level with the rate of rise mapped on top. Note that this data is not deseasoned.*

We can see a clear trend in sea level, constantly rising at a calculated average of 3.13 millimeters per year. Trends in acceleration are less clear, but calculations on the data reveal a mean acceleration of about 0.075 millimeters per year squared. These results are within the margin of error of the results determined by WCRP Group, who report a linear trend of 3.1 ± 0.3 millimeters per year and an acceleration of 0.1 ± 0.03 millimeters per year squared.[6] These values are calculated by fitting a polynomial model with least squares regression to the data and interpreting its coefficients as trend and acceleration:

```
# compute linear trend
coeffs = np.polyfit(time, gmsl, 1)
trend = coeffs[0]
print('sea level rise rate (mm/year):', trend)

# compute acceleration
coeffs_quad = np.polyfit(time, gmsl, 2)
acceleration = 2 * coeffs_quad[0]
print('acceleration (mm/year^2):', acceleration)
```

Lines 22–30 of simple_sealevel.py

The global mean sea level has been increasing since at least 1900, and the rate at which it is increasing appears to be accelerating. The primary reasons for this rise are thermal expansion and ice mass loss. As the ocean's temperature increases due to global warming, it expands, which can lead to a rise in sea surface height. According to Frederiske et al., this combined with the "increased ice-mass loss from Greenland" are the main reason for the acceleration in sea level rise. Before 1970, rises in sea level were due mostly to ice mass loss alone.[7]

[6] WCRP Group, *Earth Syst. Sci. Data*
[7] Frederikse, "The causes of sea-level rise since 1900", 393–394.

2.2 Regional Variation and Seasonal Changes

Anny Cazenave in "Contemporary Sea Level Rise" attributes about 30% of observed rate of rise to ocean thermal expansion and 55% to land ice melt (glaciers and ice sheets). The paper goes on to suggest that regional variations in sea level rise can be mostly attributed to "nonuniform changes in ocean thermal expansion". It was not possible to document these regional changes precisely before the use of satellite altimetry, since there was no way to effectively collect measurements all across the global ocean.[8]

The aforementioned "nonuniform changes" are caused by spatial differences in temperature and salinity, which affect seawater density and therefore sea surface height. These variations are known as "steric variations". If caused solely by temperature, they are called "thermosteric" or, if caused solely by salinity, "halosteric". Changes in density alter the volume of seawater, which can cause the sea level to rise without adding any new mass. These anomalies are not distributed uniformly across the globe, and therefore the sea level does not rise uniformly across the globe. Regions with strong ocean warming, like the western Pacific, tend to see above-average sea level rise, while regions with salinity increases can experience reduced rises.[9]

These kinds of variations are what we're looking for when analyzing *Simple Gridded Sea Surface Height*. We want to be able to identify regions that experience higher or lower sea level rise than average, and verify the reasons that this may be. As previously mentioned, the data is marked with its location, so it is possible to see regional trends in the context of the global trend. First, we plot the mean data to verify with our findings from the *Global Mean Sea Level* dataset. We would expect to see the same trend here as we did in **Figure 1**. This is done with the following code block:

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[8] Cazenave, "Contemporary Sea Level Rise", 145–149.
[9] Cazenave, "Contemporary Sea Level Rise", 162–165.

```
# compute global mean sea level
weights = np.cos(np.deg2rad(ds.lat))
weights.name = “weights”
gmsl = sla.weighted(weights).mean(dim=[“lat”, “lon”])
gmsl = gmsl * 1000 # convert to mm
gmsl = gmsl.compute()
```

Lines 59–64 of gridded_analysis.py

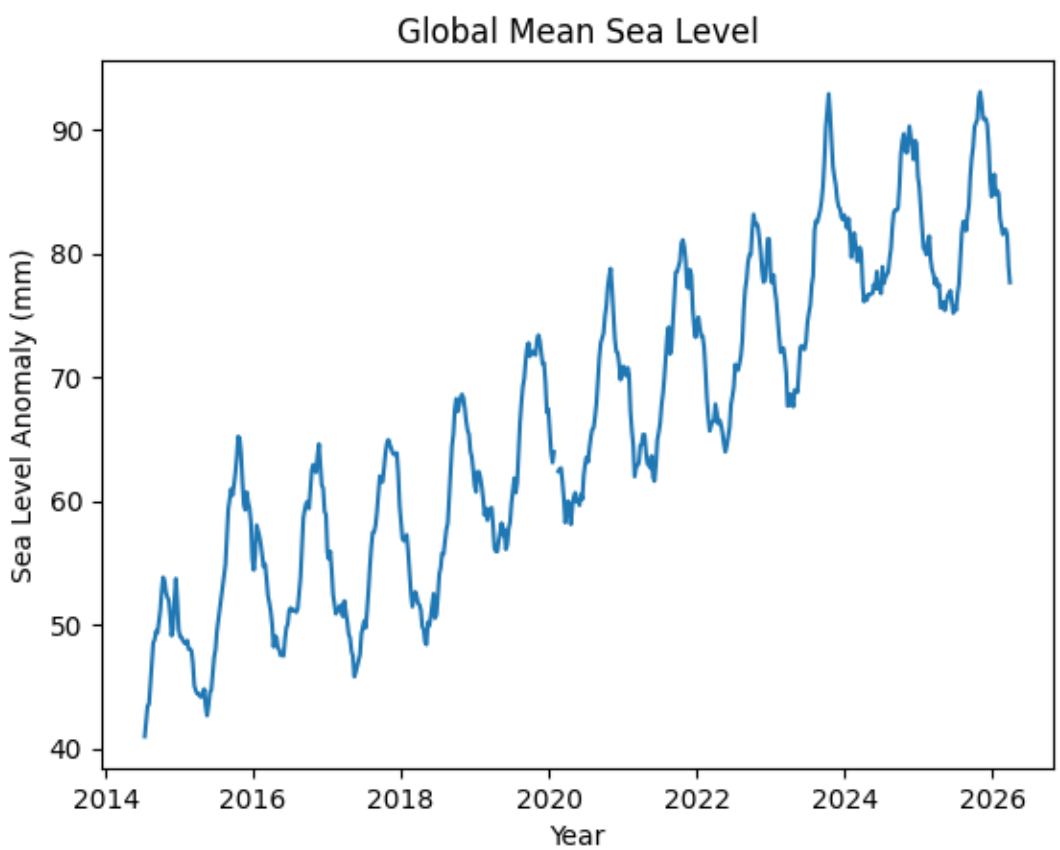

Figure 2: *Raw plot of global mean sea level from gridded data.*

This data was calculated by taking an area-weighted average over the globe. *Simple Gridded Sea Surface Height* uses 360 latitude points and 720 longitude points. However, the grid cells are not all of equal area. Near the equator, cells are larger, and closer to the poles, cells get

smaller. Therefore, using an area-weighted average is necessary over a simple mean. We can use the formula for the surface area of a sphere to get a good approximation for the weights needed for the averaging.

At first glance, the data in **Figure 2** does not seem to align with what we saw in **Figure 1**. However, because the data here is from a smaller timespan, the effects of seasonal variation are much more pronounced. Seasonal variations in sea level are caused by "global mass redistribution" between the oceans, atmosphere, and the continental hydrosphere. The water stored on land in soil, snow, rivers, and glaciers varies seasonally, and this water is exchanged with the ocean, resulting in a "seasonal signal" in sea level. Additionally, seasonal variations in atmospheric mass also redistribute water. The trends reported by Dill and Dobslaw align with what we see here in *Simple Gridded Sea Surface Height*.[10]

To get a better understanding of the general trend, we must remove the seasonal cycle from the data, or "deseason" it. Deseasoning is done by grouping all values by a certain timestamp, such as the month or day of year, and computing the mean. The code for this project also smooths the data slightly by using a rolling average, to reduce noise:

```
# remove seasonal cycle from xarray DataArray
def deseason(da, freq="dayofyear", smooth=90):
     valid = {"month", "dayofyear", "season"}
     if freq not in valid:
          raise ValueError(f"freq must be one of {valid}")

     grouped = da.groupby(f"time.{freq}")
     anomaly = grouped – grouped.mean("time")
```

Lines 42–57 of gridded_analysis.py

[10] Dill, "Seasonal variations in global mean sea level", 801–803.

Though satellite altimetry measurements are typically very low noise, noise can still be introduced into the system in many ways. This noise This noise can significantly affect sea surface height measurements, especially "poleward of the 20º latitudes".[11] Typically, this noise is removed with "low-pass filtering or least-squares curve fitting", but other, more complex methods are also being introduced, such as an Empirical Mode Decomposition based process which "adaptively projects a signal on a basis of empirical AM/FM functions called Intrinsic Modulation Functions (IMFs)". This, when applied to the data, can efficiently denoise satellite altimetry data and help us detect trends.[12]

Our program does not use any of these advanced methods to denoise the data. Because of the large quantity of data, noise in any one area does not have a significant effect. Therefore, the code simply applies a moving average after deseasoning:

```
     # apply smoothing
     if smooth is not None:
          anomaly = anomaly.rolling(time=smooth, center=True,\
min_periods=smooth // 2).mean()

     return anomaly
```

Lines 53–57 of gridded_analysis.py

After deseasoning and reducing noise, we can get a much better picture of the trend:

[11] Xu, "The Effects of Altimeter Instrument Noise", 2229.
[12] Quilfen, "On denoising satellite altimeter measurements", 875.

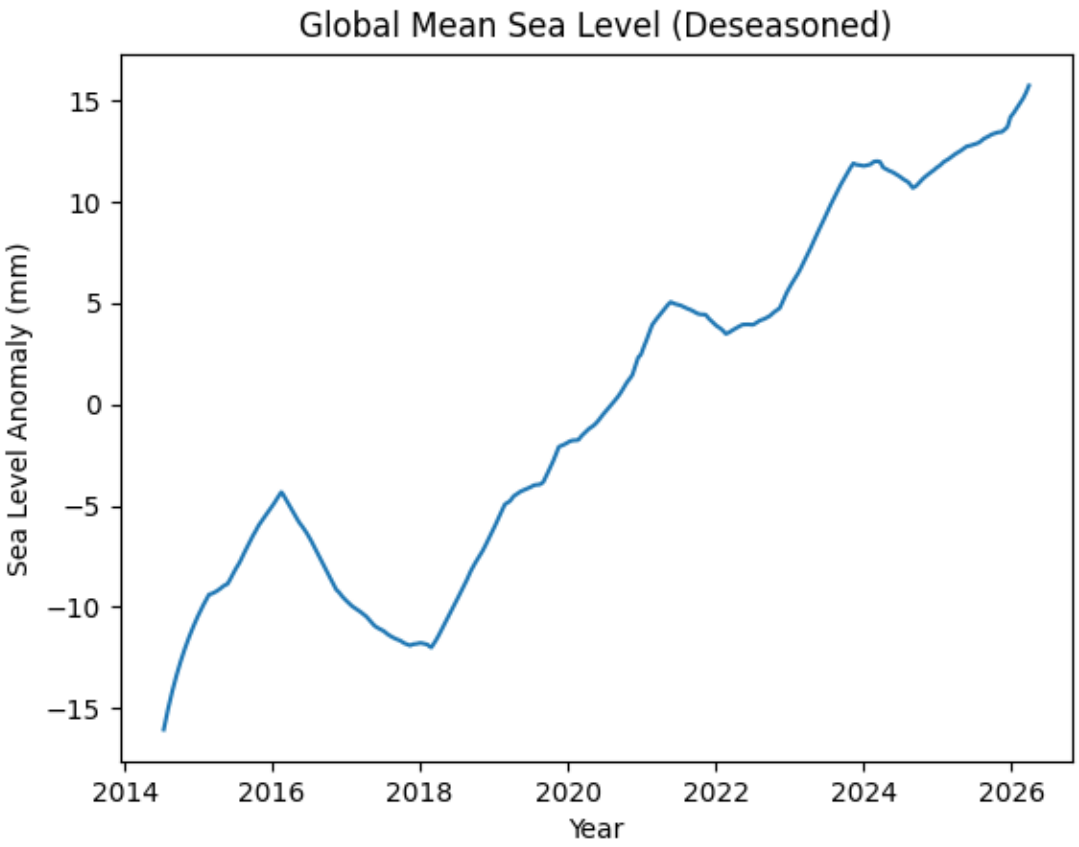

Figure 3: *Deseasoned and smoothed mean sea surface height data from the gridded dataset*

It is much easier to see the trend from **Figure 1** in the desesasoned and smoothed data than it was in **Figure 2** before these operations were performed. This shows both the significant effects that seasonal variation has on sea surface height, and the importance of understanding your data when performing analysis. When we calculate the linear trend on this data, we get a value of 3.326 millimeters per year, which is comparable to the 3.13 millimeters per year we calculated from *Global Mean Sea Level Trend*.

Now, let's take advantage of the main feature of the *Simple Gridded Sea Surface Height* dataset and analyze regional deviations from the global trend. First, we define regions for analysis by creating a bounding box:

```
# regional analysis
regions = { # define regions to analyze
     "Equatorial Pacific": dict(lat=slice(-10, 10),\
lon=slice(150, -90)),
     "North Atlantic": dict(lat=slice(0, 60), lon=slice(-80,\
0)),
     "Indian Ocean": dict(lat=slice(-30, 30), lon=slice(40,\
120)),
}
```

Lines 108–113 of gridded_analysis.py

These bounds are extracted from the full dataset so we can perform analysis with only this data. Then, we calculate regional means in the same way as the global mean, taking a weighted mean within the bounds.

```
# calculate means
regional_means = {}
for name, bounds in regions.items():
     subset = sla.sel(**bounds)
     regional_means[name] = subset.weighted(weights).mean(dim=[\
"lat", "lon"]) * 1000
```

Lines 115–119 of gridded_analysis.py

When we plot this data, we can begin to see the regional variation in sea surface height:

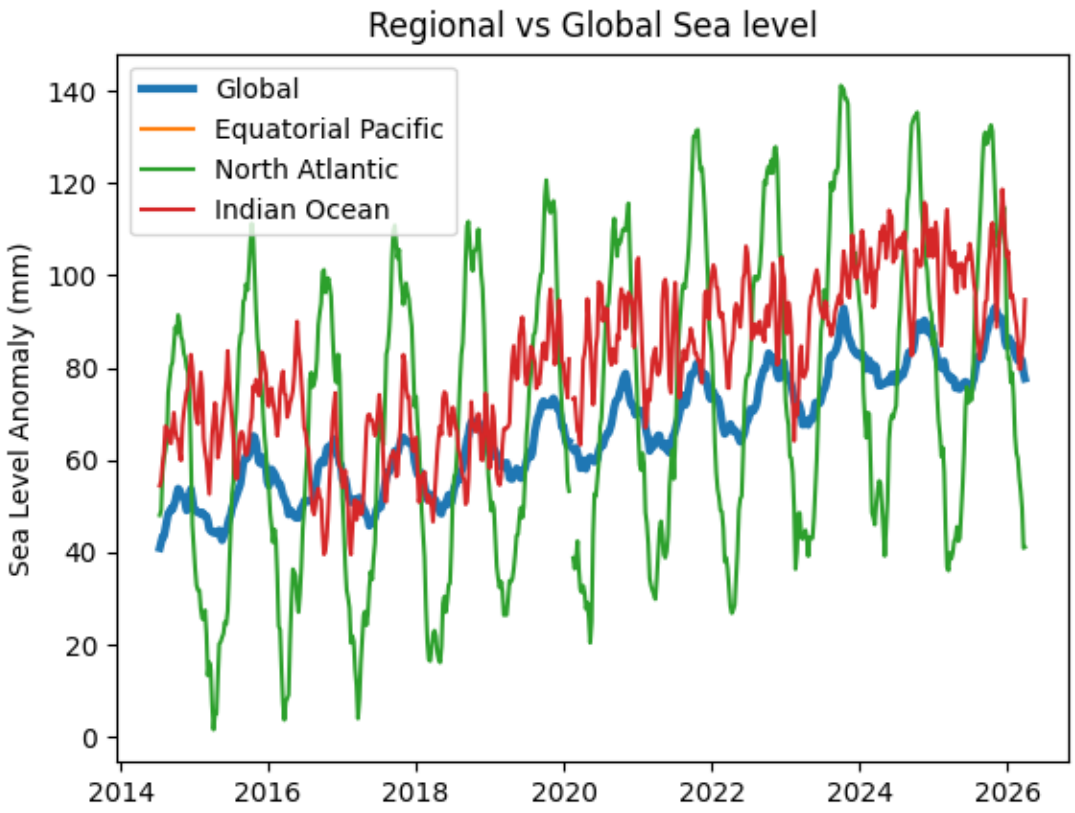

Figure 4: *Sea surface height data for various regions, with the global trend for reference.*

As we can see, this data also exhibits seasonal trends. One thing to note is the unique seasonal variation in different regions. It's important to recalculate deseasoning for each region, as seasonal signals can vary widely based on regions (weaker variation in the tropics, stronger signal in higher latitudes).

The strongest variation we can see in the plot is in the North Atlantic. Seasonal variation of sea surface height in the North Atlantic is primarily caused by "the seasonal wind stress cycle" and steric effects, similar to what was discussed before.[13] The largest variability is seen near the

[13] Ferry, "Seasonal SSH variability in the North Atlantic", 6307–6315

Gulf Stream, where warm seawater from the south flows north to relatively cooler waters. This causes a "cliff effect", where sea surface height is higher than normal. As the strength of the Gulf Stream is affected by seasonal changes, so is the sea surface height in this area. [14]

The line for the equatorial pacific closely follows the global trend, to the point that it is not even visible on the plot. Seasonal variations in the equatorial pacific are very low compared to those in other regions, and so we see very little effect on sea surface height. One exception to this is the South China Sea (which is found along the equator). Here, seasonal variation is more pronounced due to "a pileup of seawater driven by the northeasterly monsoon" in the winter causing a rise, and because of eastward "jet-like" flows in the fall.[15]

The Indian Ocean's seasonal variability is due to a similar reason. This region's upper ocean circulation is "mainly driven by the seasonal reversal winds of the monsoon". The Indian Monsoon Current seasonally redistributes heat from the Arabian Sea and Bay of Bengal, causing a "net positive heat gain across the air-sea interface" over the course of the year. This affects the thermal structure in the upper ocean. These seasonal changes in temperature and salinity alter the density of seawater, leading to expansion and contraction of the water column, directly affecting sea level. In addition, the monsoon also forces Kelvin and Rossby waves, which can have a significant effect on ocean circulation. These waves affect not just the Indian Ocean, but "can propagate through long distances and can affect ocean circulations in remote regions far from their origin", causing seasonal variations throughout the global ocean.[16]

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[14] Kelly, "Seasonal Variations of SSH in the Gulf Stream", 313.
[15] Chung-Ru, "Seasonal variability of SSH in the South China Sea", 13982.
[16] Salim, "SSH Variability in the Tropical Indian Ocean", 679–683.

2.3 Regional Variation in the General Trend

Now, let's remove the seasonal signal from this data and apply smoothing so we can clearly see the general trend. This is done the same way as before, by grouping values by timestamp, computing a mean, and taking a moving average. As discussed previously, we take care to deseason each region individually to eliminate the specific effects seen in those regions:

```
# deseason
regional_deseasoned = {
     name: deseason(xr.DataArray(data.values, coords=[ds.time],\
dims=["lat", "lon"]) * 1000
}
```

Lines 121–125 of gridded_analysis.py

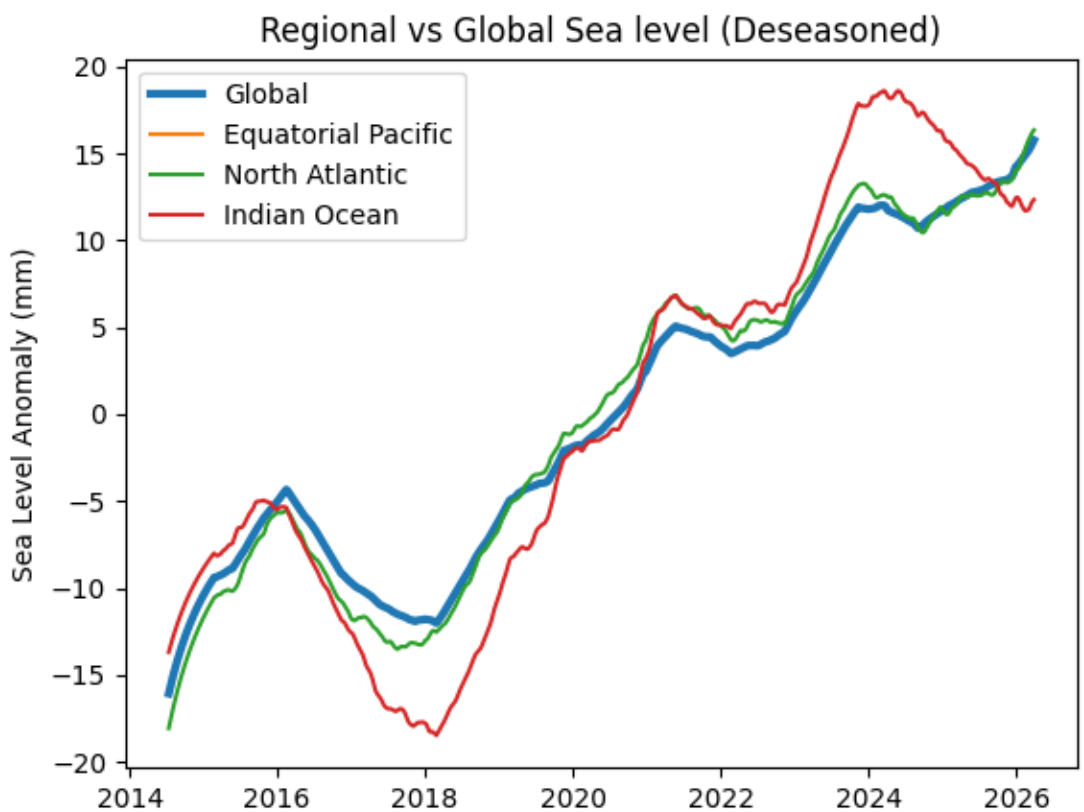

Figure 5: *Deseasoned sea surface height data for different regions*

Now we begin to see the same trend as before. Note that the regional variation here is much less than what we saw in the "seasoned" data.

Much of the regional variation on the large-scale trend is thanks to variations in temperature. Ocean heat content is strongly connected to sea surface height, "particularly … on monthly to interannual time scales". This is essentially the same steric effect we have been discussing when it comes to seasonal effects. Warmer water leads to thermal expansion, leading to lower density in the seawater, resulting in higher sea surface height. Therefore, increase in ocean heat content generally produces positive sea surface height anomalies. Of course, temperature alone is not the only contributing factor to sea surface height variation. Salinity changes can also alter density, and this effect "dominate[s] seawater density changes at colder temperatures".[17]

The line that deviates farthest from the general trend on the plot is that of the Indian Ocean. When we keep in mind the connection to temperature, this makes sense. The subtropical Indian Ocean dipole has been responsible for variability in upper ocean heat content on a large time scale. One study found that the subtropical Indian Ocean dipole is "associated with a prominent southwest-northeast sea level anomaly". Though the result is somewhat similar, this is distinct from the effects of monsoon winds. The authors of the study note, however, that "the SLA pattern associated with the subtropical Indian Ocean dipole mode remains unknown". Though the data shows that the sea level anomaly is definitely connected to this dipole, there is less confidence on the exact pattern.[18]

This high variability in the Indian Ocean has been visible historically as well. A 2000 study on sea surface temperature across different regions found "larger warming trends in the

[17] Fasullo, "On the Relationship between Regional OHC and SSH", 9195–9199.
[18] Zhang, "Variability of Sea Level and OHC in the Indian Ocean", 7228–7230.

Indian and Southern Hemisphere basins". As we have seen, temperature is one of the largest factors in sea level rise. Though the data in this study was taken between 1945 and 1995, if we extrapolate forward, we can see that it agrees with our findings.

3.0 MAPPING THE DATA

Perhaps the strongest rebuttal of the "bathtub model" is a visual representation of the data on a map. Using *Simple Gridded Sea Surface Height*, we can take all the grids and project them onto a world map, highlighting the variation between sea surface height anomaly in different regions.

The map the program creates is a simple subtraction of the sea level in the latest measurements from the sea level in a particular past year. Because the time axis in *Simple Gridded Sea Surface Height* is irregular, there is some math involved in calculating the correct number of time steps to take for each region to get an accurate representation. Then, means are calculated so we can calculate the change in sea level.

```
# calculate per-pixel sea level change using mean of first and\
last n_years
def compute_endpoint_trend(sla, n_years=5, use_means=True):
     if use_means:
          timesteps_per_year = len(sla.time) / (
               (sla.time[-1] – sla.time[0]).values/np.timedelta\
(1, 'D') / 365.25
          )
          n = int(n_years * timesteps_per_year)

          start = sla.isel(time=slice(0, n)).mean("time")
          end = sla.isel(time=slice(-n, None)).mean("time")
     else:
          start = sla.isel(time=0)
          end = sla.isel(time=-1)

     return (end – start) * 100 # convert to cm
```

Lines 153–167 of gridded_analysis.py

We then compute the change in sea level while creating the map, and use a color scale to show the differences in sea level across the map. We use a Robinson projection for the map. It is important to scale properly:

```
# plot the map
def plot_sea_level_change_map(sla, n_years=5, vmax=None, \
use_means=True):
     change = compute_endpoint_trend(sla,n_years=n_years)
\.compute()

     # calculate scale
     if vmax is None:
          vmax = float(np.nanpercentile(np.abs(change.values),\
95))
     fig, ax = plt.subplots(
          figsize=(16, 8),
          subplot_kw={"projection": ccrs.Robinson()}
     )

     img = ax.pcolormesh(
          sla.lon, sla.lat, change
          transform = ccrs.PlateCarree(),
          cmap="RdBu_r",
          norm=mcolors.TwoSlopeNorm(vmin=-vmax, vcenter=0, \
vmax=vmax),
          shading="auto",
     )

     ax.add_feature(cfeature.LAND, facecolor="lightgray", \
zorder=1)
     ax.add_feature(cfeature.COASTLINE, linewidth=0.5, zorder=2)
     ax.gridlines(draw_labels=True, linewidth=0.3, alpha=0.5)

     start_year = int(sla.time[0].dt.year)
     end_year = int(sla.time[-1].dt.year)
```

Lines 169–195 of gridded_analysis.py

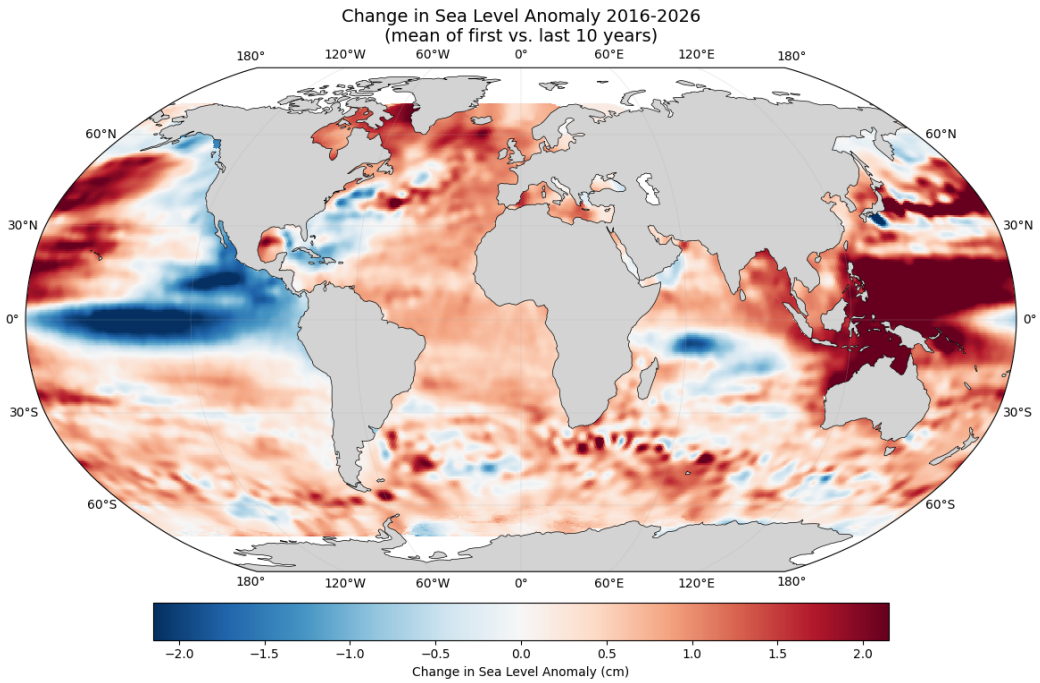

Figure 6: A map showing the sea level anomaly across the globe over the last 10 years.

From this map, it is clear that change in sea surface height is highly dependent on region. There is a high level of variability, with some places experiencing over 2 centimeters of sea level rise and others actually experiencing a drop of 2 centimeters. One area of note is the equatorial Pacific. We can see a huge area experiencing a massive drop in sea level anomaly. This is mostly a result of the El Niño-Southern Oscillation effect, which causes anomalies in sea surface temperature and pressure, which in turn affect sea surface height. This effect has been particularly strong in the past 10 years, leading to the extreme effect we see in **Figure 6**.[19]

[19] Xuan, "ENSO Sea Level Pressure Anomalies", 8860.

4.0 CONCLUSIONS

Sea surface height is a variable that can be affected by a wide variety of factors. Therefore, it makes sense that different regions experience sea level change differently. We have discussed these reasons at length, but one thing I would like to highlight is the level of detail that can be extracted from satellite altimetry data. All the calculations, plots, and visualizations in this report were generated from just 2 datasets. Satellite altimetry is a remarkable tool in developing our understanding of the Earth's global ocean system.

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